

MURANG'A UNIVERSITY OF TECHNOLOGY

Continuous Assessment Test

AMS 617: Statistical Computing

Instructions

This work is to be done individually.

Submit to: *anapapa@mut.ac.ke* by the end of the day on 25th March, 2025.

Maximum number of pages allowed is 15

Background Information

The housing dataset contains factors like house area, bedrooms, furnished, nearness to main road, etc. that may be used to predict housing price.

The data contains 13 variables and 545 observations.

The variables are:

Variable	Description
price	Price of the house
area	Area of the house
bedrooms	Number of house bedrooms
bathrooms	Number of bathrooms
stories	Number of house stories
mainroad	Whether connected to main road
guestroom	Whether house has a guest room
basement	Whether house has a basement
hotwaterheating	Whether house has a hot water heater
airconditioning	Whether house has air conditioning
parking	Number of house parkings
prefarea	House in a preferred area?
furnishingstatus	Furnishing status of the house

Question One – Use R

- Generate appropriate graphs to present the data
- Perform descriptive analysis of the dataset
- Formulate five objectives, perform tests of hypothesis and discuss the results

Question Two– Use Python

- a) Generate appropriate graphs to present the data
- b) Perform descriptive analysis of the dataset
- c) Formulate five objectives, perform tests of hypothesis and discuss the results

Question Three

Compare the code and output of the analysis in Question One and Question Two explaining which software you will recommend.